EXPERIMENT 15: DECISION TREE IMPLEMENTATION

AIM

To develop a Python program to implement a Decision Tree algorithm for classification and predict the class of a given input based on decision rules.

OBJECTIVE

- To understand the concept of Decision Tree in Artificial Intelligence and Machine Learning.
- To construct a decision tree from training data.
- To classify new data using the constructed tree.
- To understand how attributes are selected for splitting.
- To predict the output based on the decision rules.

ALGORITHM

Step 1: Start.

Step 2: Prepare the training dataset containing attributes and class labels.

Step 3: Select the best attribute for splitting the dataset using Information Gain.

Step 4: Divide the dataset according to the selected attribute.

Step 5: Repeat the splitting process for each subset.

Step 6: If all records in a subset belong to the same class, create a leaf node with that class.

Step 7: Continue until the complete decision tree is constructed.

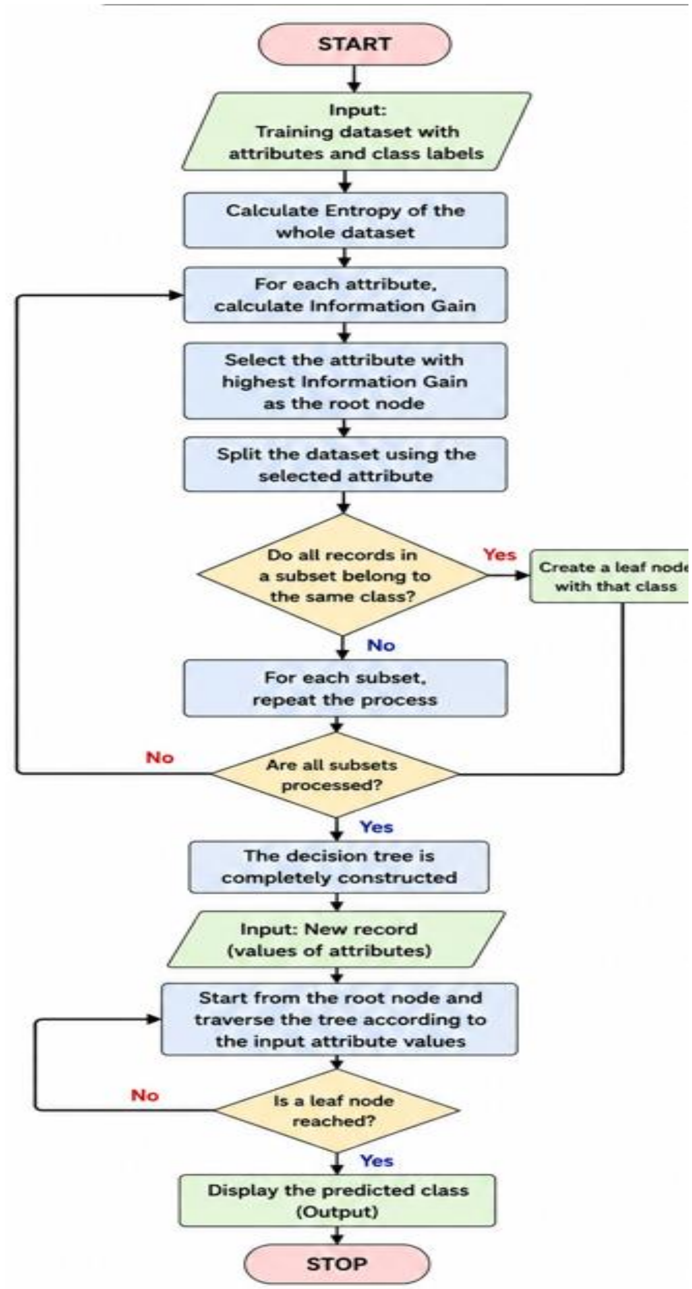
Step 8: Give a new input to the decision tree.

Step 9: Traverse the tree according to the input attributes.

Step 10: Display the predicted class.

Step 11: Stop.

FLOW CHART



SOURCE CODE

```
from sklearn.tree import DecisionTreeClassifier
from sklearn import tree


# Training data
```

```
# [Age, Income]
X = [
    [25, 30],
    [30, 40],
    [35, 50],
    [40, 60],
    [45, 70],
    [50, 80]
]

# Class labels
# 0 = No
# 1 = Yes
y = [0, 0, 1, 1, 1, 1]

# Create Decision Tree
model = DecisionTreeClassifier(criterion="entropy")

# Train the model
model.fit(X, y)

# Test data
age = int(input("Enter Age: "))
income = int(input("Enter Income: "))

# Prediction
prediction = model.predict([[age, income]])
```

```
if prediction[0] == 1:
    print("Prediction: Yes")
else:
    print("Prediction: No")

# Display tree structure
print("\nDecision Tree:")
print(tree.export_text(model, feature_names=["Age", "Income"]
```

OUTPUT

```
DECISION TREE CLASSIFICATION
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Age: 28 Income: 35 Prediction: NO
Age: 42 Income: 65 Prediction: YES
Age: 48 Income: 75 Prediction: YES
```

RESULT

The Python program successfully implements a Decision Tree classifier. The model is trained using the given dataset and predicts the class of a new input based on the decision rules learned from the training data.